

User Requirements Specification

Speech and LLM Quality Lab

User requirements for a locally installable application that provides traceable evaluation of speech-to-text and language-model responses.

Item	Decision
Document version	2.2 · 6 October 2026 · English edition of baseline 2.1
Responsibility	Pascal Cabrel Nague · personal development and learning project
Product milestone	V0.1 local peer beta · target date 13 October 2026
Usage	Each tester installs the application on their own computer.
Status	Revised plan; implementation and acceptance are not yet claimed.
Language	English interface, documentation and project discussions; French speech test cases remain the evaluation domain.

Purpose and authority

The tool examines the quality of speech data, ASR transcripts and LLM responses. It is an independent personal project with no external client. A model error is something to investigate; an incorrect score or unintended data disclosure is a defect in the tool.

This revision separates a beta that can be tested within seven days from the later full evaluation product. The restart concerns goals, priorities and understanding. Existing code will be reviewed and reused where suitable, rather than discarded without review.

Collaboration and learning objective

During active work sessions, the AI assistant handles implementation proposals, integration, tests and documentation. Pascal makes the engineering decisions, runs measurements on his computer and understands the problem, data, metrics, architecture, security boundaries and results. Line-by-line coding lessons are not a prerequisite.

For each important decision, explain the need, alternatives, rationale, evidence and limitations. Pascal should then be able to explain the decision in his own words. A schedule does not start unattended background work. From this edition onward, collaboration is in English, with corrections to Pascal’s English when needed.

1 Beta goals and deliverables

1.1 Value for testers

A tester starts the tool without editing source code, examines an approved French example or their own short recording, sees errors and missing results, reviews a response case and can export feedback. The beta tests the usability of the workflow; it is not general evidence of model quality.

Goal	V0.1 acceptance
Z1 Usable ASR workflow	At least one real local ASR run; display audio, hypothesis and an optional checked reference. WER/CER require a valid reference.
Z2 Understandable evaluation	Show normalization, denominators, case status and the distinction between data errors and model errors.
Z3 Learn LLM evaluation	Six curated illustrative response cases, one per behavior group; four criteria and a justified rating.
Z4 Data control	Storage location, temporary and saved data, export and deletion are accessible from the interface.
Z5 Peer feedback	Test guide and manually exportable feedback form. Target: 3–5 testers; responses cannot be guaranteed.

1.2 Required scope

Local startup and storage, model weights that can be installed in advance, and an offline example package. Recording in a desktop browser or import of a valid WAV file; validation before inference; one fixed ASR profile.

At least 12 approved real audio–text pairs, two per linguistic category; development only, not a final test benchmark.

Case view, WER/CER, manual critical-error flags and locally persistent rating revisions after explicit saving. JSON/CSV/HTML reports with provenance labels; safe feedback export that excludes user content by default.

1.3 Explicitly outside V0.1

No public hosting, central accounts, multi-user server or cloud inference. No live local LLM generation, model training, fully automated end-to-end benchmark or claimed regression approval. No smartphone installation package; responsive presentation is prepared, but LAN access is not enabled.

The previous targets of 120 audio cases, 60 LLM cases and 30 test pairs remain goals for V1.0. They are explicitly not release conditions for 13 October 2026.

2 Usage and system boundaries

Role	Responsibility and access
Pascal	Product owner, engineering decision-maker and developer with AI assistance. Receives explicitly shared feedback and selected diagnostic cases.
Local tester	Processes their files on their own computer and decides what to save, delete and share. No automatic disclosure to Pascal.
Other speakers	Recordings require documented permission for use; no implicit permission for publication or training.

2.1 Core workflows

Use case	Expected workflow
UC01 Getting started	Install, read the brief privacy notice, check the storage location and open the example package.
UC02 Evaluate ASR	Record or import audio; validate its format; transcribe; check the reference; understand raw and normalized error rates.
UC03 Explain errors	Filter number, homophone, elision, negation, identifier or colloquial-language cases; listen to and annotate discrepancies.
UC04 Review a response	Examine the question, sources, response and expected behavior; save four ratings and a justification.
UC05 Manage data	List saved personal cases, deliberately trigger a full local data export, or delete cases and their derivatives.
UC06 Provide feedback	Rate tasks, describe errors, inspect the preview and voluntarily share feedback or a selected diagnostic case manually.

2.2 Operating environment

The beta primarily targets Windows 11 x64 with Python 3.11. Ubuntu x64 support is desirable and may be claimed only after an actual installation test. macOS, ARM and smartphones are not promised for V0.1. The release manifest states supported platforms.

The planning reference is the previously documented Windows computer with a Celeron N5100, 32 GB RAM and Intel UHD graphics. Confirm this profile again on J1. A GPU is not required; real-time transcription is not promised. Report measured resources for other devices.

Installation and an explicit model download may require internet access. Once prepared, the core workflow must work without outbound network access. Redistribution rights must be established for all sample recordings; permission for local use does not automatically permit redistribution.

3 Functional requirements and migration

B = MUST for beta V0.1; V = MUST for the later full product V1.0; S = SHOULD. Version 1.0 IDs are retained. The scopes below replace the previous implicit six-week deadlines. Unmet V requirements remain open, not passed. J1–J7 identify project days 1–7.

ID	Beta	Requirement and scope boundary
F01	B	Validate schema, audio, types, IDs and safe paths before inference; reject invalid files with an understandable error.
F02	B	Store versioned references and ratings with pending/approved/unclear status; no silent changes to completed results.
F03	B	Review audio, reference and hypothesis together; filters do not alter data.
F04	V	Compare two ASR configurations on the same frozen dataset. Beta: one real adapter and saved examples.
F05	B/V	Beta: correct raw/normalized WER and CER. Automatic slot error rate is deferred to V1.0; critical errors are initially marked manually.
F06	B	Show categories, severity and denominators; flag small samples.
F07	B	Show loading time, inference time, timeouts and cancellations separately; no failure disappears from reporting.
F08	B/V	Beta cases remain dev data; no unjustified holdout claims. A locked, grouped test split belongs to V1.0.
F09	V	Reproducible document retrieval. The beta shows fixed associated sources without pretending to perform retrieval.
F10	V	Real LLM generation with saved inputs and model versions; outside the beta.
F11	B	Four criteria, justification, revision and rating coverage; label illustrative cases explicitly.
F12	V	Compare two prompt variants on the same frozen dataset.
F13	V	Paired ASR/reference branches with fresh retrieval. The beta may show only explicitly labeled illustrations.
F14	V	Compatibility and regression gates. The beta shows “No comparable run yet”, not an invented success.
F15	B	Dashboard with horizontal navigation, status messages, filters, case view, data management and feedback.
F16	B	Safe JSON/CSV/HTML exports with provenance, denominators, coverage and limitations.
F17	B/V	Beta: atomic saving per case and explicit partial status after cancellation. Cross-run resume/cache is deferred to V1.0.
F18	B	Short personal recording or WAV import for exploratory tests; never automatically add it to a test split.
F19	B	Local data overview, understandable saving choices and complete case-level deletion.
F20	B	Manually exportable peer feedback with preview; no automatic transmission.

4 Data and quality

ID	Beta V0.1	Later V1.0
D01	12 real approved audio cases: 6 categories $\times$ 2, all dev. Simple tone fixtures do not count as speech.	120 cases: 15 dev and 5 test per category; 90/30 overall.
D02	6 illustrative LLM cases, one per behavior group, dev only. Do not derive a model score from them.	60 cases: 5 dev and 5 test per group; 30 test cases linked to audio.
D03	At least 3 fictional support texts with stable source IDs for the review workflow.	12–20 original documents and defined gold passages.
D04	Provenance, license, speaker permission and redistribution approval for each bundled item.	Same rule for all further data and models.
D05	Traceable version, hash, provenance type, annotation and date; do not pass the reference text to ASR.	Also dataset freeze, leakage checks and complete paired mapping.
ID	Required beta quality criterion	
N01	Identical saved predictions and evaluation rules yield identical scores. Model inference may vary.	
N02	Invalid inputs, incomplete runs and write errors remain visible. No false success messages.	
N03	1,000 saved fixture cases on the reference device: initial display ≤ 5 s after startup; p95 ≤ 2 s over 20 filter changes. Cold start/model loading measured separately.	
N04	CPU operation; hardware pilot before fixing the model. Support profile: at least 8 GB RAM and 3 GB free storage, to be confirmed on J2.	
N05	Local operation, no automatic user-data transfers, no secrets in the repository or default export.	
N06	Separate UI, storage, validation, adapters and evaluation; automated tests cover critical rules.	
N07	Installation without code changes, understandable errors, labeled controls, visible keyboard focus and limited animation.	
N08	Every result identifies dataset, configuration, status and denominator; demo and real inference are distinguishable.	

User feedback and exploratory speech tests do not form a scientific holdout. Additional speakers broaden observations but do not establish robustness across accents or domains. An incorrect reference is a data problem and must not be reported as a model error.

5 Security needs and control over data

Each tester has their own local installation. There is no central application server or user account with Pascal. Pascal maintains a separate local collection of voluntarily received diagnostic cases. The application is accessible only on its own computer. This operating boundary is a release condition, not an interchangeable convenience setting.

ID	MUST requirement
S01	No access from the LAN or internet. Default configuration must not enable a public listener, tunnel or external inference service.
S02	Before the first input, clearly explain processing, storage location, retention, export and deletion. Recording one's own voice is optional.
S03	Data directories outside source code; no automatic cloud sync, telemetry or upload. Run under a normal user account.
S04	Explicit saving is required. Temporary files are cleaned up at session end or expiry; saved content can be deleted completely by case.
S05	Reject invalid, oversized or manipulated files. File names and contents must not affect files outside the data root.
S06	User text and model text remain data. Do not execute HTML, scripts, spreadsheet formulas or commands, or automatically open links.
S07	By default, diagnostics and feedback contain no audio, transcripts, usernames, absolute paths or secrets.
S08	Version and verify the installation package, dependencies and model provenance. No uncontrolled updates during a test.
S09	Limit file size, runtime, queue length and storage use to prevent accidental overload.
S10	Sharing requires deliberate export and manual transfer. Privacy defects block beta release.

5.1 What Pascal receives

Nothing by default. A tester may create a feedback package after previewing its contents. It contains task status, ratings, release ID and an optional comment. Default feedback excludes speech data. F21 additionally permits a separately confirmed diagnostic-case export with optional audio. Shared feedback is stored separately on Pascal's computer and deleted no later than 30 days after receipt; abstract improvement tickets without personal data may remain.

5.2 Limitations

A voice or free-text field may contain personal data; a random case code does not make it anonymous. Only fictional support requests are requested. The tool does not protect against malware, administrator access or other processes on a compromised computer. File deletion does not guarantee forensic erasure; user-created exports and system backups are outside the application's control.

6 Acceptance and testing with classmates

Gate	Release condition
G1 Core functions	Real ASR inference on the reference device; 12 checked distributable audio cases and 6 illustrative response cases; core tasks work.
G2 Measurement	Hand-calculated WER/CER examples pass; missing references show N/A; failures and small denominators remain visible.
G3 Privacy and security	Local listener, outbound traffic, file, export, quota and deletion tests pass; no unresolved critical finding.
G4 Installation	Fresh installation on a second supported Windows computer without a development environment; model preparation and offline startup documented.
G5 Handover	Release package, instructions, brief privacy notice, test form, known limitations and evidence matrix are available.

All B requirements, S01–S12 and beta-relevant D/N requirements need evidence. Missing evidence does not count as a pass. Release records include commit, package hash, device profile, test date and Pascal’s decision. Historical test counts do not automatically demonstrate revised requirements.

6.1 Test protocol for one person

Attempt installation and first startup; record duration, obstacles and help needed.
 Open an example, listen to the audio and explain a number or negation error.
 Optionally record a fictional request; confirm the reference and interpret WER/CER.
 Review a response case with its sources and find the saved review after restarting.
 Delete a personal case, then check the list, report and audio access.
 Rate usefulness, clarity and usability from 1 to 5 each; identify the biggest problem and one suggested improvement.

6.2 Targets and limits of interpretation

Target 3–5 voluntary responses. Aim for at least three people to finish the core workflow without changing code; count assistance and abandonment. Report fewer responses openly; they do not retrospectively invalidate a technically correct release. Do not derive statistical claims from this small sample.

A security incident or data loss stops distribution and further user-data testing. A poor ASR score does not, provided it is correctly calculated and clearly presented. Cosmetic issues may remain with an open ticket.

7 Seven days to a usable version

Start: 6 October 2026. Target: V0.1 handover on 13 October 2026. The seven work periods run from 6–12 October; 13 October is the release and delivery day. This is a committed planning target, not a promise independent of tests and tester availability.

Day	Outcome	Pascal's contribution and understanding
J1 · 6 Oct	Review specifications and existing code; confirm storage path and hardware.	Explain audience, scope, requirements/design specifications and threat model; approach testers.
J2 · 7 Oct	Local startup, data management, file validation, installation and model pilot.	Understand storage types, trust boundaries and CPU/runtime measurement.
J3 · 8 Oct	Real audio-to-transcript workflow, reference checking and 12 approved examples.	Explain reference errors, WER/CER and normalization using three cases.
J4 · 9 Oct	Case analysis, six response cases and persistent rating revisions.	Distinguish groundedness, clarification, abstention and rating coverage.
J5 · 10 Oct	Exports, case sharing, incoming-package validation, deletion, resource limits and security tests.	Explain the data lifecycle, input attacks and safe outputs.
J6 · 11 Oct	Second device, installation, offline testing, usability and fixes.	Review test evidence and distinguish software quality from model quality.
J7 · 12 Oct	Freeze the release candidate; finalize instructions and known limitations.	Review the release matrix and give a five-minute engineering demonstration.
13 Oct	Check gates; distribute the package and begin collecting feedback.	Document release approval; accept no unresolved critical security vulnerability.

7.1 Effort and fallback

Planning assumption: 30–40 hours of technical implementation and testing, including reuse, plus 6–8 hours from Pascal for data, device measurements and engineering decisions. These are estimates. Reassess remaining effort on J2; a reserve of about 20% is included in the technical budget.

If delayed, first drop Linux approval, optional visual views and unrestricted result imports. Real ASR processing, correct metrics, local data control and security gates remain mandatory. If time is insufficient, clearly label a restricted preview; it does not count as accepted V0.1.

8 Risks changes and further development

Risk	Mitigation
CPU too slow	Test a small multilingual ASR model on J2; use prepared real examples for usability; no real-time guarantee.
Installation fails for friends	Prioritize Windows x64, tested locks, a startup script and an early second installation; explicitly verify the minimum profile.
Missing references or permissions	Use only 12 carefully checked distributable core cases; replace unclear cases and require license evidence.
Data enters Git or feedback	Separate storage root, export allowlist, redacted logs and negative tests.
Scope growth	Defer new requests until after V0.1 unless they close a security or acceptance gap.
Feedback mistaken for a benchmark	Keep the beta exploratory; define study questions and a holdout before V1.0.

8.1 Revisions and traceability

CR-001 records the shift to a personal project and replacement of the old schedule. CR-002 defines the local peer beta. CR-003 adds S01–S10 and F19–F20. CR-004 establishes engineering-focused collaboration. Historical IDs are not reassigned; deferred requirements retain their meaning and open status.

The revised User Requirements and Technical Design Specifications apply together as version 2.2. Older editions are historical records, not the current release basis. Changes to metrics, normalization, datasets or rubrics require new version identifiers and recalculation where necessary.

8.2 Next product stages

V0.2 responds to peer feedback: improve installation, assess more speakers and data sources, and enable a second ASR comparison. V1.0 completes F04, F09–F10, F12–F14, F17 and D01–D03. Measure and replan scope and dates after the beta.

8.3 Terms for engineering explanations

Term	Meaning in this project
Requirement / solution	The User Requirements Specification states the need and verifiable outcome; the Technical Design Specification defines implementation and evidence.
Baseline / holdout	A baseline is a recorded comparison point; a holdout is a dataset reserved for final evaluation.
Data quality / model quality	Incorrect references or files are data problems; incorrect model responses despite correct inputs are model problems.
Coverage / pass rate	Coverage measures completeness; pass rate requires a defined passing rule and a visible denominator.
Threat model / gate	A threat model identifies possible attacks and boundaries; a gate is a verifiable release condition.

9 Voluntary diagnostic cases for improvement

CR-005 expands the feedback scope. Pascal needs reproducible failure cases to improve the application systematically. Local installation remains unchanged. An additional deliberate export enables access to selected data; there is no remote access or automatic collection.

ID	MUST for V0.1
F21	“Share this case with Pascal”: select one case, display fields and purpose, edit content, select audio separately, then confirm or cancel export.
F22	Pascal can safely validate received packages, open them as unverified diagnostic cases and link them to an improvement ticket. No automatic inclusion in training data or a test benchmark.
S11	Record the consent decision for the specific export. Audio is not preselected. Show recipient, included data, purpose and deletion deadline before export.
S12	Keep received cases outside Git; restrict access, retain for 30 days from receipt and delete on request using the sharing code. Track derived copies too.

9.1 Purpose and minimum data

Usability issues require task status and a comment. Depending on the error, ASR diagnosis needs audio, hypothesis, checked reference, model configuration and timings. LLM evaluation needs the question, sources, response and justified rating. Share only the selected case, never the whole workspace. A report without audio may be insufficient for acoustic diagnosis; state this limitation.

9.2 Transfer and use

V0.1 creates a package for manual transfer through a non-public channel agreed before testing. The tester decides whether to send it. Export does not mean receipt. Raw data must not appear in GitHub issues or public links. The test notice explains transfer outside the application and copies held there.

The purpose is diagnosis and improvement of this personal project. This permission does not authorize publication, model training or onward sharing. Longer retention or later use for another purpose requires a separate explicit agreement. After deletion, retain only abstract tickets without personal data or newly written synthetic tests.

9.3 Acceptance and schedule impact

G3/G5 also cover B17–B19: correct selection and preview, safe incoming-package validation, deletion and reproduction evidence. Implement export and intake on J5 and test on two devices on J6. The target remains 13 October; reassess the added effort on J2. Defer visual extras and Ubuntu support before these MUST functions.

10 Technologies and competency objectives

The User Requirements Specification requires understandable, maintainable and testable components. Binding language choices appear in Technical Design section 14. This overview defines learning and delivery goals without requiring additional frontend or cloud stacks.

Area	Planned means	Competency and evidence
Application logic and ASR	Python	Separate modules; explain input/output contracts and failure modes; demonstrate a real ASR case.
Interface and reports	Python/Streamlit, HTML, CSS	Assess workflows, safe output, responsive presentation and keyboard operation.
Data and configuration	JSON/JSONL, CSV, TOML	Explain schema, version, provenance, retention and validation. These are data/configuration formats.
Installation and operation	PowerShell; Bash only for supported Linux	Explain environment, dependencies, startup, permissions and reproducible installation.
Quality and testing	Python/pytest; Git; Markdown	Maintain acceptance criteria, negative tests, evidence and change decisions.
Later extensions	Python; SQL or JavaScript/TypeScript only after a new architecture decision	Not mandatory for the beta. A database, separate frontend and cloud operation are separate learning and project goals.

10.1 Competency evidence as a separate deliverable

The third document, “Competency Portfolio”, records existing foundations, evidenced practice, skills to consolidate and planned learning goals by category. It separates engineering understanding, practical execution and coding independence. A feature implemented with AI assistance does not automatically prove personal mastery.

ML Evaluation, ML Engineering, MLOps, Data Engineering, Data Analytics, AI Efficiency and AI Application Engineering are considered separately. Security and requirements management are cross-cutting competencies; completing this beta alone does not establish security or architecture expertise.

10.2 Document set

User Requirements 2.2 defines needs and success; Technical Design 2.2 defines the solution and verification; Competency Portfolio 1.1 records Pascal’s learning status and verifiable progress. CR-006 adds technology mapping and competency evidence. Original requirement IDs remain; F21/F22 and S11/S12 are additions. CR-007 introduces English throughout and Times New Roman; the French evaluation data domain is preserved.